Countable Borel Relations, Invariant Measures, and Graphings

Equivalence Relation Preliminaries

First, we collect some definitions.

Definition:

- A relation R on a set X is a set of ordered pairs $R \subseteq X^2$. We will write xRy to denote $(x,y) \in R$. We also define the *sections*

$$R_x = \{y : (x,y) \in R\}$$
$$R^y = \{x : (x,y) \in R\}$$

  and the inverse relation

$$R^{-1} = \{(y,x) : (x,y) \in R\}.$$

- If X is equipped with a Borel structure, then the relation R (i.e., equivalence relations, partial orders, total orders, etc.) is called Borel if $R \subseteq X^2$ is Borel.
- We will let $\operatorname{dom}(f)$, $\operatorname{ran}(f)$, and $\operatorname{graph}(f)$ denote the domain, range, and graph of f respectively.
- A graph G with vertex set X is a non-reflexive, symmetric relation on X. The neighbors of $x \in X$ in the graph G are the set of all $y \in X$ such that $(x,y) \in G$. The cardinality of the set of neighbors of X is called the degree of x, denoted $d_G(x)$.
- A path in G is a finite sequence of vertices $(x_0, x_1, \dots, x_n)$ where $(x_i, x_{i+1}) \in G$, and $x_i \neq x_j$ whenever $i \neq j$ except for possibly $x_n = x_0$. The equivalence classes of the relation xEy if and only if there exists a path from x to y defines the connected components of the graph. If a graph has one connected component, we say the graph is connected.
- If $x_n = x_0$ with $n \geq 3$, then we call the path a cycle. A graph that has no cycles is called acyclic, and a connected acyclic graph is called a tree. A rooted tree is a tree with a distinguished vertex x_0 called its root.
- If X is a standard Borel space and E is a Borel equivalence relation on X, then we say the relation is countable if $[x]_E$ is countable for each $x \in X$.
- If $\Gamma \curvearrowright X$ is a countable group acting on X such that Γ preserves the Borel structure of X, then we define the orbit equivalence relation to be $xE_\Gamma^X y$ if and only if there exists g such that $g \cdot x = y$.
- The orbit space X/Γ is defined to be the quotient X/E_Γ^X.
- If E is a countable Borel relation, then the *full group* of E, denoted $[E]$, is the set of Borel automorphisms whose graphs are contained in E. The set of partial Borel automorphisms is denoted $[[E]]$; that is,

$$[[E]] = \{\phi\colon A \to B : \phi \text{ is a bijection}, A, B \subseteq X \text{ are Borel}, \operatorname{graph}(\phi) \subseteq E\}.$$

- We call a countable equivalence relation E on X *aperiodic* if every equivalence class $[x]_E$ is infinite.
- Given Borel equivalence relations E and F on X and Y respectively, we say E is *Borel reducible* to F, denoted $E \leq_B F$, if there is a Borel map $f\colon X \to Y$ such that xEy if and only if $f(x)Ff(y)$.

Example: If X^Γ denote the space of functions of Γ onto X, then the left shift action of Γ on X^Γ is defined by $\gamma \cdot p(\delta) = p\left(\gamma^{-1}\delta\right)$.

If G is a standard Borel group and $\Gamma \subseteq G$ is a countable discrete subgroup (also known as a lattice), then the action of Γ onto G by left translation, taking $\gamma \cdot g = \gamma g$, yields the quotient space G/Γ given by the set of *right* cosets of Γ.

Example: The following are examples of countable Borel equivalence relations.

(i) If $X = 2^{\mathbb{N}}$, then the *eventual equality* relation says that xE_0y if and only if there exists m such that for all $n \geq m$, we have $x_n = y_n$. The tail equivalence relation says that xE_ty if and only if there exist m and k such that for all n, we have $x_{m+n} = y_{k+n}$.

(ii) If $X = \mathbb{R}$, then the *Vitali relation* says that xE_vy if and only if $x - y \in \mathbb{Q}$.

(iii) If $X = \mathbb{R}_+$, then the *commensurability relation* says that xE_cy if and only if $x/y \in \mathbb{Q}$.

(iv) If $k \geq 2$ and X is the space of subshifts of $k^{\mathbb{Z}}$, where a subshift is a nonempty closed subset of $k^{\mathbb{Z}}$ invariant under the shift map $(S(x))_i = x_{i-1}$.

This is a compact subspace of the hyperspace of compact subsets of $k^{\mathbb{Z}}$, so it is a compact and metrizable space. Define two subshifts to be isomorphic if there is a homeomorphism between the two closed subsets that commutes with the shift, and define an equivalence relation as such.

Theorem (Feldman–Moore)**:** Let E be a countable Borel equivalence relation on X. Then, there is a countable group Γ and a Borel action of Γ on X such that $E_\Gamma^X = E$.

Proposition (Marker Lemma)**:** Let E be an aperiodic countable Borel equivalence relation on X. Then, there is a sequence $\{S_n\}_{n\geq 1}$ of Borel sets $S_n \subseteq X$ such that

(i) $S_0 \supseteq S_1 \supseteq S_2 \supseteq \cdots$;

(ii) $\bigcap_{n\geq 1} S_n = \emptyset$;

(iii) every S_n is a complete section for E (i.e., it has nonempty intersection with each equivalence class of E).

Such a sequence is called a *vanishing sequence of markers*.

Proof. Via some results from descriptive set theory, we may assume that the equivalence relation E lives on the set $X = 2^{\mathbb{N}}$. Then, we let $s_n(x)$ be the lexicographically smallest $s \in 2^n$ such that $|[x]_E \cap \mathcal{N}_s| = \infty$, where $\mathcal{N}_s$ denotes the cylinder set in X corresponding to s.

Define $x \in A_n$ if and only if $x|_n = s_n(x)$. Then, we have that $\{A_n\}_{n\geq 1}$ is a decreasing sequence of Borel sets which intersects each class of E at infinitely many points. Then, $A = \bigcap_{n\in\mathbb{N}} A_n$ intersects each class of E in at most one point, so that $B_n = A_n \setminus A$ is the vanishing sequence of markers as desired. □

Invariant Measures

Definition: If E is a countable Borel equivalence relation on X, then we say μ is E-invariant if for all $f \in [E]$, we have $f_*\mu = \mu$, where $f_*\mu$ is the pushforward measure, defined by

$$f_*\mu(A) = \mu\left(f^{-1}(A)\right).$$

Proposition: The following are equivalent:

(a) μ is E-invariant;

(b) μ is Γ-invariant for any countable group Γ such that $E = E_\Gamma^X$;

(c) μ is Γ-invariant for some Γ such that $E = E_\Gamma^X$;

(d) for all $\phi \in [[E]]$, we have $\mu(\mathrm{dom}(\phi)) = \mu(\mathrm{ran}(\phi))$.

If μ is invariant under E, then we may define a measure $\overline{\mu}$ on E by taking

$$\overline{\mu}(A) = \int |A_x| \, d\mu(x),$$

where again,

$$A_x = \{y : (x, y) \in A\}.$$

It is also possible to define

$$\overline{\mu}'(A) = \int |A^y|\,d\mu(y),$$

but since μ is invariant, $\overline{\mu} = \overline{\mu}'$. This follows from the fact that

$$E = \bigcup_{i\in\mathbb{N}} \operatorname{graph}(g_i)$$

for some collection of relation-preserving automorphisms $g_i \in [E]$. If $A \subseteq E$ is any Borel subset, then

$$A = \bigcup_{i\in\mathbb{N}} (\operatorname{graph}(g_i) \cap A),$$

so there are $f_i \in [[E]]$ such that $\operatorname{graph}(f_i) = \operatorname{graph}(g_i) \cap A$. Therefore, we may write any $A \subseteq E$ as a countable disjoint union of graphs of functions $f \in [[E]]$. Therefore, it suffices to show that $\overline{\mu}(\operatorname{graph}(f)) = \overline{\mu}'(\operatorname{graph}(f))$ for any $f \in [[E]]$. If $C = \operatorname{dom}(f)$ and $D = \operatorname{ran}(f)$, then $\overline{\mu}(\operatorname{graph}(f)) = \mu(C)$ and $\overline{\mu}'(\operatorname{graph}(f)) = \mu(D)$, so by the definition of $[[E]]$, we are done.

Now, we may define an equivalence relation $\widetilde{E}$ on E by taking $(x, y)E(z, w)$ if and only if xEz, or equivalently, $xEyEzEw$.

Proposition: The invariant measure M on E is $\widetilde{E}$-invariant.

Proof. Let Γ act on X such that $E = E_\Gamma^X$. Then, if we define the action $(g, h) \cdot (x, y) = (g \cdot x, h \cdot y)$, then we have that $E_{\Gamma^2}^E = \widetilde{E}$, so it suffices to show that the action preserves M.

For this, if $A \subseteq E$ is Borel, we have

$$M((g,h)\cdot A) = \int \left|\{(g\cdot x, h\cdot y) : (x,y) \in A\}\right| d\mu(x).$$

Setting $B = (g, h)\cdot A$, then we have $y \in B_x$ if and only if $(x, y) \in B$, which holds if and only if $\left(g^{-1} \cdot x, h^{-1} \cdot y\right) \in A$, which holds if and only if $h^{-1} \cdot y \in A_{g^{-1}\cdot x}$. Therefore, $B_x = h \cdot A_{g^{-1}\cdot x}$, so that

$$\begin{aligned} M((g,h)\cdot A) &= \int \left|A_{g^{-1}\cdot x}\right| d\mu(x) \\ &= \int |A_x|\,d\mu(x) \\ &= M(A), \end{aligned}$$

where we use the fact that μ is invariant under Γ. $\square$

Graphings and Cost

Definition: If $E \subseteq X^2$ is a countable Borel equivalence relation, then a (Borel) *graphing* of E is a graph G such that the connected components of G are the equivalence classes of E. We say that G generates E.

Definition: A Levitt graph (L-graph) is a countable family $[[E]] \supseteq \Phi = \{\varphi_i\}_{i\in I}$ of partial Borel isomorphisms $\varphi_i : A_i \to B_i$, where A_i, B_i are Borel subsets of X.

We call Φ *finite* if I is finite. We say Φ is an Levitt graphing (L-graphing) if Φ generates E in the sense that xEy if and only if $x = y$ or there is a finite set $i_1, \dots, i_k \in I$ and $\varepsilon_1, \dots, \varepsilon_k \in \{1, -1\}$ such that $x = \varphi_{i_1}^{\varepsilon_1} \cdots \varphi_{i_k}^{\varepsilon_k}(y)$.

For every L-graph Φ, there is an associated graph G_Φ that generates the same equivalence relation. We say $xG_\Phi y$ if and only if $x \neq y$ and there exists i such that $\varphi_i(x) = y$ or $\varphi_i(y) = x$.

Conversely, for every graph G, there is an L-graph Φ_G such that $G = G_{\Phi_G}$. This follows from fixing a Borel (total) ordering $<$ on X, and finding a family of partial Borel functions $\{f_i\}_{i\in\mathbb{N}}$ with disjoint graphs such that for all $x \in X$, we have

$$\{(x,y) : xGy \text{ and } x < y\} = \bigcup_{i \in \mathbb{N}} \operatorname{graph}(f_i).$$

Since the f_i are countable-to-one, we have $\{g_{i,j}\}_{j \in K_i}$ a family of partial Borel automorphisms with disjoint graphs such that

$$\operatorname{graph}(f_i)^{-1} = \bigcup_{j \in K_i} \operatorname{graph}\big(g_{i,j}$$

(Recall that we define the inverse of a relation to be the coordinates flipped.) Define $\Phi_G = \{g_{i,j}\}_{i,j}$.

Considering the case of an E-invariant μ, if G is a graphing of $E|_A$ for some conull E-invariant Borel subset A, then we say G is a graphing of E almost everywhere. Similarly, we may define an L-graphing almost everywhere.

Now, we will discuss the theory of cost.

> **Definition:** Let E be a countable Borel equivalence relation on X, and let μ be an E-invariant measure with corresponding measure on E given by M. If $G \subseteq E$ is a subset, then we define its *cost* to be
>
> $$C_\mu(G) = \frac{1}{2} M(G).$$

Observe that $0 \leq C_\mu(G) \leq \infty$, and $0 < C_\mu(G)$ if G is a graphing of E provided that it's not the case that almost every E-class is a singleton. This follows from the fact that G is a complete section for the relation $\widetilde{E}$ on E, meaning that $M(G) > 0$.

Now, if $\Phi = \{\varphi_i\}_{i \in I}$ is an L-graph, define its cost by taking

$$C_\mu(\Phi) = \sum_{i \in I} \mu(\operatorname{dom}(\varphi_i)) = \sum_{i \in I} \mu(\operatorname{ran}(\varphi_i)).$$

Then, we have

$$C_\mu(\Phi) = \frac{1}{2} \int \sum_{i \in I} \Big(\chi_{A_i}(x) + \chi_{B_i}(x)\Big)\, d\mu(x),$$

where we have $A_i = \operatorname{dom}(\varphi_i)$ and $B_i = \operatorname{ran}(\varphi_i)$.

References

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